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An automated pipeline for LOFAR very-long baseline interferometry

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Software

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Summary

The Very-Long Baseline Interferometry (VLBI) Pipeline for the International Low-Frequency ARray Telescope (PILoT) is an automated data reduction pipeline that produces calibrated radio data suitable for sub-second resolution imaging. It is a tool that facilitates calibrator and source selection, self-calibration of data, and both postage-stamp and widefield imaging.

While a diverse ecosystem of processing and imaging tools exists for the LOFAR telescope, none of those tools have been designed with high-resolution imaging in mind. As a result, data reduction with the International LOFAR Telescope is a manual and error-prone process. Furthermore, owing to the distributed nature of software development in the LOFAR community, all of these tools have been developed with different input and output conventions.

PILoT aims to incorporate these diverse software tools into a unified framework, making VLBI imaging with LOFAR accessible to a larger group of astronomers. Special care has been placed on ensuring that PILoT is interoperable and reusable, and that all of its software components are controlled through a consistent framework and that intermediate steps of the pipeline can be consistently and safely resumed in the event of intermediate failure. Since typical observations done with the International LOFAR Telescope result in datasets consisting of terabytes of data, the pipeline has been designed to integrate with job schedulers common in High-Performance Computing (HPC) clusters. This minimises manual intervention and optimises the use of available computing resources.

Statement of need

The International Low-Frequency ARray Telescope (ILT) (van Haarlem et al., 2013) currently comprises 38 Dutch stations and 14 international stations located in partner countries across Europe. It is a radio telescope operating at radio frequencies under 240 MHz with a sensitivity of up to 3 orders of magnitude better than previous telescopes operating at comparable frequencies. By combining data from all stations with baselines of up to 1980 km, the ILT is effectively a continent-sized telescope which is able to image astronomical radio sources at sub-arcsecond resolution (Leah K. Morabito et al., 2025; Varenius et al., 2015).

The VLBI Pipeline for the International LOFAR Telescope (PILoT) is an implementation of a data reduction pipeline which was designed to exploit the full imaging power of the ILT (L. K. Morabito et al., 2022). PILoT addresses several critical issues the original reference implementation had to face:

- The original pipeline was implemented in an obsolete framework, which makes it difficult to impossible to ensure that it would be functional on modern computing infrastructure. In contrast, PILoT is implemented in the Common Workflow Language (CWL; Crusoe et al. (2022)), which is an actively maintained framework in widespread use. This ensures that the pipeline will be logically consistent and maintainable in the long term.
- The original pipeline did not support modern scheduling systems such as Slurm or TORQUE. The implementation in CWL allowed for optimisation of PILoT for the workflow runner toil (Vivian et al., 2017), providing native support for these schedulers. This reduces the runtimes of the pipeline by orders of magnitudes as individual processing jobs can automatically be distributed to available nodes.

In addition, PILoT includes expanded functionality featuring implementations of state-of-the-art advances in imaging techniques such as improvements in imaging resolution (F. Sweijen et al., 2022; Ye et al., 2024), source selection (J. M. G. H. J. de Jong et al., 2024; F. Sweijen et al., 2022), and wide-field imaging techniques (J. M. G. H. J. de Jong et al., 2025; F. Sweijen et al., 2022).

State of the field

Many software tools have been developed in order to process LOFAR data. Some of these are designed specifically for LOFAR; these allow for correcting for various instrumental and ionospheric effects (de Gasperin et al., 2019) and calibrating for directional effects using the data obtained by the Dutch stations using pipelines such as DDF-pipeline (Hardcastle et al., 2019; Tasse et al., 2021) or Rapthor (Rafferty et al., 2021). This class also includes the LOFAR Initial Calibration (LINC) pipeline (Drabent et al., 2022), which performs image calibration using the Dutch stations. Other tools are more widely applicable in radio astronomy, such as DP3 (Dijkema et al., 2023), WSclean (Offringa et al., 2014), AOflagger (Offringa, 2010), which are developed by the ILT host institute ASTRON, as well as various codebases developed by researchers in the LOFAR community such as the DDF-pipeline (Tasse et al., 2021) for direction-dependent calibration, LOFAR facet-selfcal (R. van Weeren, 2022; R. J. van Weeren et al., 2021) for self-calibration, and the LOFAR Helpers (J. de Jong et al., 2026; J. M. G. H. J. de Jong et al., 2022) auxiliary library.

As these tools were designed either as general-purpose tools in radio astronomy or for calibration of the Dutch LOFAR array, none of these tools are particularly suited to high-resolution imaging. Instead, PILoT was developed to form a natural bridge to the full ILT and is designed from the ground up to produce high-resolution images of radio sources.

As a result, PILoT embeds the software tools above into a single cohesive framework and provides several complementary modes with the aim of being a modular and fully automated imaging pipeline. It is able to determine when intermediate results are no longer needed, and disposes of intermediate results once they are no longer required.

Software design

PILoT's principle requirements are that the processing of LOFAR data must be portable, reproducible, and scalable, while keeping a uniform API. It uses CWL to ensure that its workflows are logically consistent. Furthermore, the pipeline is optimised to work with the CWL runner toil, which is the most mature CWL implementation available. Individual processing steps can easily be resumed should the execution of a mode be interrupted for any reason.

CWL also allowed for PILoT's API to be agnostic of the underlying software tools, which are written in Python or C++. It also allowed for an intuitive and straigtforward integration with queueing systems common in HPC systems.

For convenience, the pipeline has been structured into several workflows, which are grouped according to different modes. The pipeline provides the following main modes of operation:

Postage stamp imaging

The pipeline supports single-source imaging, in which the calibration is performed on an in-field calibrator to correct for direction-independent phases and delays from the international stations. PILoT selects the best calibrator based on the images and solutions from all performed self-calibration cycles. Following this, multiple imaging targets can be specified at once; the pipeline performs self-calibration and imaging on each target (with a resolution of 0.3 arcseconds) in parallel. The data products are the calibrated data and images and the phase solutions in a H5parm format[1].

Wide-field imaging

The pipeline is capable of intermediate (1–2 arcseconds) and high-resolution (sub-arcsecond) imaging. Before imaging the pipeline removes radio sources from observation data beyond a square field of view of 6.25 degrees^2 centred on the imaging target, if given a 6 arcsecond image generated by the DDF-pipeline. High-resolution imaging supports a resolution of 0.6 or 0.3 arcseconds, of which the former reduces the imaging time by a factor of 4 compared to the latter. Intermediate resolution imaging speeds up the imaging time by a factor of 16 compared to the 0.3 arcsecond imaging. The data products are FITS formatted images of the stated resolution.

Research impact statement

Since its initial release, PILoT has been integrated into the FLoCs LOFAR containers (Frits Sweijen et al., 2025). FLoCs are the standard software containers used for data processing in the LOFAR astronomy community.

In addition, the majority of the 15 LOFAR2.0 Large Programs have requested to use PILoT, as they require high-resolution imaging in order to achieve their science goals. These projects fall in four different themes of research which are described in section 4 of reference (Leah K. Morabito et al., 2025).

Furthermore, ASTRON has committed to support development of PILoT as part of the LOFAR Enhanced Network for Sharp Surveys (LENSS). As part of this project PILoT will be incorporated into the ASTRON's operational environment.

Finally, PILoT is used to test and benchmark the computing resources of the SKA Regional Centre (SRC) in the United Kingdom as part of the LOFARINT demonstrator case.

AI usage disclosure

No generative AI was used in the development of PILoT, the writing of this manuscript, or the preparation of supporting materials.

[1] The H5parm format is described in appendix C of (de Gasperin et al., 2019).

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The scripts developed for PILoT make use of the ASTROPY (Astropy Collaboration et al., 2022), casacore (Casacore Team, 2019), LoSoTo (de Gasperin et al., 2024), numpy (Harris et al., 2020), pandas (McKinney, 2010), and PyBDSF (Mohan & Rafferty, 2015) libraries.

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